NAAMA SKITAL

SOFTWARE ENGINEER | MACHINE LEARNING | COMPUTER VISION

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PROFILE

Computer Science Honors graduate (GPA 92.17) focused on applied ML, computer vision, NLP and model evaluation. Built reproducible classification experiments with PyTorch, CLIP and scikit-learn, including multimodal image/OCR fusion, calibration and class-imbalance analysis. Also brings production-facing Python/Linux engineering experience.

TECHNICAL SKILLS

Languages: Python, C++, C, Java, SQL **Core:** Algorithms, Data Structures, OOP, Debugging, Testing
AI / Data: PyTorch, scikit-learn, NumPy, pandas, CLIP, NLP, Computer Vision, Model Evaluation
Systems / Tools: Linux, Git, ROS2, DDS, Raspberry Pi, Sensor Synchronization, Jupyter

PROFESSIONAL EXPERIENCE

Software Engineer | LMY *2025–2026*

- Developed Python software for a Linux-based UAV payload collecting LiDAR and navigation sensor data for 3D mapping workflows.
- Worked with ROS2/DDS messaging, sensor timing and synchronization, Raspberry Pi services, and data acquisition pipelines.
- Built validation and diagnostic scripts; investigated asynchronous data, communication delays, and timing inconsistencies.
- Supported post-flight processing and troubleshooting across embedded and ground-side software components.

SELECTED PROJECTS

Multimodal Advertisement Moderation | Python, PyTorch, CLIP, OCR GitHub

- Built and evaluated a four-class image moderation pipeline; compared frozen CLIP, partial fine-tuning, and image/OCR fusion.
- Best fusion model achieved **91.00% accuracy** and **0.6551 macro-F1**; analyzed imbalance and calibration.

Context-Aware Sarcasm Detection | NLP, MiniLM, scikit-learn GitHub

- Compared TF-IDF/logistic regression and sentence-embedding baselines across comment-only and context-aware inputs.
- Achieved **72.53% accuracy** with the comment-only TF-IDF baseline; examined the effect of context.

Topological Data Analysis | Python, GUDHI, Persistent Homology GitHub

- Studied Vietoris–Rips filtrations, H_0/H_1 persistence, diagrams and stability under perturbations on synthetic point clouds.

Software and Algorithms | C++, Java C++ / Java

- Built nine C++ coursework implementations covering OOP, templates, STL and generic containers; implemented Java Bayesian network inference with variable elimination.

Online Purchase Prediction | scikit-learn, Classification GitHub

- Evaluated classifiers on 12,330 sessions; investigated imbalanced labels, threshold tuning, feature importance, PCA and SMOTE.

EDUCATION

B.Sc. Computer Science | Ariel University *Nov 2023 – Aug 2026*
Honors Program (AI / Data Science) · GPA: **92.17** · Rector's Excellence (first year)
Relevant coursework: Algorithms, Data Structures, Operating Systems, Computer Networks, ML, Deep Learning, NLP and Computer Vision.

NATIONAL SERVICE, TEACHING & VOLUNTEERING

National Service (2 years): Supported at-risk youth at Dolev residential school and a Jerusalem high school for students who had left mainstream education; taught mathematics.
Mathematics Teaching: Prepared at-risk high-school students for the 4-unit mathematics matriculation examination.
Community Volunteering: Bnei Akiva youth counselor, including mentoring children with special needs; volunteered with children at Schneider Children's Medical Center.